
CNN-recurrent connectivity hybrids for visual working memory tasks

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Abstract

We are interested in creating a model that can solve an array of visual working memory tasks of escalating complexity. To do this, we propose three CNN-recurrent connectivity hybrid models, and test them on a variety of delay-match-to-sample (DMTS) tasks, which require the model to maintain a sample image in working memory and compare it with a later test image under delays and transformations to determine whether they are repeated by assigning a label. We discover that these models exhibit a few biologically plausible traits, including a diminishing performance in labeling test images as delay rises. Models also exhibit an incapacity to scale to tasks with variable delay, and perform better with distractor image delays than with empty delays, which contrasts standard working memory models for animals. We also explore the regions most informative to model labeling by implementing Grad-CAM, introduced in (Selvaraju et al., 2016), and find that regions closer to image object ground-truth location are more informative for prediction.

1. Introduction

Working memory has many definitions, though is generally agreed to regard the temporary storage and manipulation of information (Baddeley, 2012; Wilhelm et al., 2013). We are interested in evaluating models' visual working memory, and identifying their working memory capacity. A suite of common tasks considered for this purpose in psychology are visual delay-match-to-sample (DMTS) tasks, where the subject is presented with a sample stimulus, and from a subsequent set of stimuli are required to identify which "matches" the sample. Because we are more interested in the fundamental question of "can these models exhibit working memory" than "what are the peculiarities of its

working memory capacity, if our models exhibit it", we reduce these tasks, so that in our experiments the subject is presented a sample image, and after delay is required to identify whether a subsequent test image is repeated.

Developing models that exhibit working memory, and evaluating their biological plausibility, has been addressed before in a variety of standard models. Recurrent neural networks (RNNs) have become popular for evaluating cognitive theories due to their ease of training, and have historically performed well on other working memory tasks, but standard RNNs are not designed to complete visual tasks, such as identifying repeated images; similarly convolutional neural networks (CNNs), while used for visual data, struggle in tasks where neural representations have longer temporal dependencies. Thus we consider introducing recurrent connectivity layers to standard CNN models, letting neurons feed their outputs back into their own inputs to create internal memory storage, potentially allowing the modified CNN to learn temporal dependencies on image tasks.

2. Related Work

The proposed DMTS tasks have been prior studied in humans and in monkeys. (Chelazzi et al., 1998; 2001) find that macaques demonstrate a high capacity to perform the proposed task when abstracted to a "good" and "bad" stimulus, whereby the monkey has to match a sample stimulus to another in a test array after delay. Interestingly, the results of Chelazzi et al.'s experiments demonstrate that macaque attention follows a "biased competition" model, whereby objects observed by the macaque compete for cortex representation and is skewed in favor of the behaviorally relevant stimulus.

Results like that of Chelazzi et al. motivate (Lorenc et al., 2021), who argues that distraction resistance is a pronounced capacity shared in animals when exhibiting working memory and is consequently distraction resistance is a necessity in any working memory model.

Training biologically-plausible RNNs for working memory tasks has been explored in different settings, like in (Soo et al., 2023), which proposes a specialized skip-connection through time to embed long temporal dependencies, though this exploration is limited and has rarely been carried to

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the visual domain. Given the natural intuition that images provide when considering how the model operates (i.e. what is the subject looking at? What is it being distracted by? How powerful are these effects?) carrying this study to a visual working memory task such as DMTS is merited. The research in developing an interpretable model in this area is limited, and though (Yang et al., 2023) propose a possible model experimenting with other simple model alternates is needed.

The cognitive science canon altogether shows the importance of object location and distraction in our working memory tasks is marked, and evaluating whether any working memory exhibited in our proposed models considers these features can be informative to the underlying mechanics of how the models store short-term memory.

3. Experiments

Our images are taken from the Imagenette, which is a subset of ImageNet, a benchmark dataset for images used in visual classification tasks (see A for further discussion). Our images are then resized to 224 by 224 pixels, which is the standard input size for most ResNet models.

We complete two DMTS tasks, which are each composed of four experiments of ascending complexity. In task 1, our delay frames are blank (i.e. 224 by 224 pixel images of 0 values), while in task 2 our delay frames are distractor images (i.e. other images from Imagenette, downscaled to 224 by 224 pixels). In experiment 1 of each task, we have a fixed number of delay frames and do not subject the test image to transformation. In experiment 2 of each task, we have a fixed number of delay frames and subject the test image to transformation. In experiment 3 of each task, we have a number of delay frames variable by trial between 1 and 25, and the test image does not have transformation. Our transformations are given by the TensorFlow functions `transforms.RandomHorizontalFlip()`, as well as a resize of our image to 320 by 320 pixels and consequent `transforms.CenterCrop()` back to a 224 by 224 pixel image.

In order to trial these tasks, we propose three CNN-recurrent connectivity hybrids. As we are performing our experiments using subsets of the ImageNet dataset, we incorporate ResNet-50 as a pretrained CNN model, so that our hybrids inherit the learned features of ResNet-50 on the broader ImageNet dataset for classification of Imagenette. Our models should then be able to classify and distinguish images without delay, and performance in our DMTS tasks should be attributable to the capacity to store information about the sample in working memory under delay.

Our first proposed model simply inputs our images, goes through all the convolutional layers of ResNet-50, then has

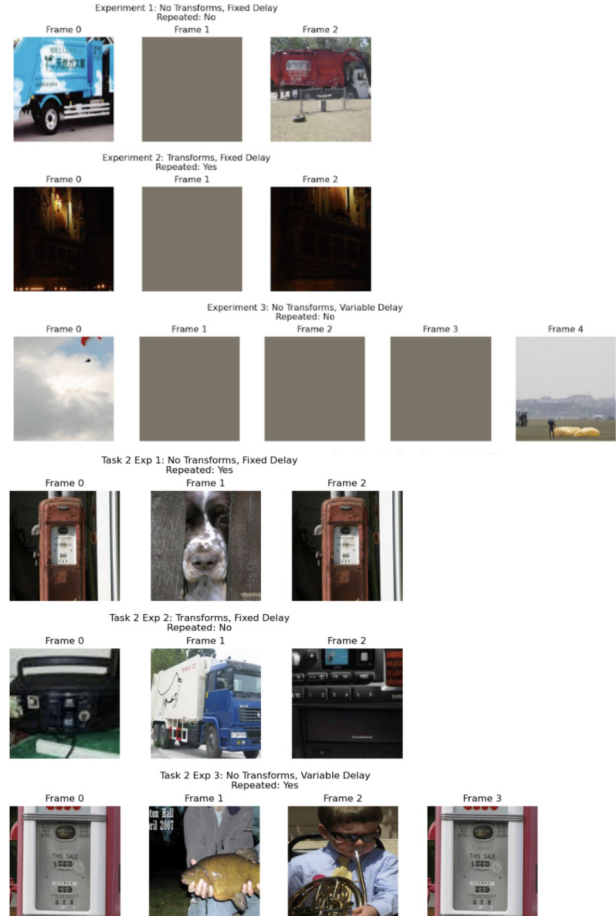


Figure 1. Example trials of our experiments in task 1 (top) and task 2 (bottom).

a recurrent connectivity layer prior to a final fully-connected layer with a labeling decision.

Our second proposed model inputs our images, goes through the initial convolution layer, and then interleaves each layer stage of ResNet-50 with a recurrent connectivity layer, prior to a fully-connected layer with a labeling decision.

Our third proposed model inputs our images and takes them through the convolutional layers of ResNet-50 separately. Then, we append a ConvLSTM layer (introduced in (Shi et al., 2015)) and feed our convolutional layer results for each image separately. Finally, we return to our fully-connected layer for a labeling decision (auxiliary information about the proposed models can be seen B).

4. Results and Analysis

We begin with experiments 1 and 2, for a fixed delay of a single frame. Our interleaved and separate hybrids perform

generally well on these experiments. It is notable that the models seem to perform better on distractor images than on empty frames. I posit this is due to empty frames being out-of-distribution relative to the average color values encountered in ImageNet; future work is needed to validate this, and to possibly implement "empty frames" with color values of the ImageNet mean instead. Behaviorally, this result contrasts what is expected by the "biased competition" model as observed in animals, as a distractor should limit the model's capacity for attention on retaining the features of the sample image.

Table 1. Validation losses for interleaved and separate hybrid on task 1 experiments 1 and 2, with fixed delay 1, by epoch 5.

MODEL	EXPERIMENT 1	EXPERIMENT 2
INTERLEAVED	0.594	0.602
SEPARATE	0.187	0.242

Table 2. Validation for interleaved and separate hybrid on task 2 experiments 1 and 2, with fixed delay 1, by epoch 5.

MODEL	EXPERIMENT 1	EXPERIMENT 2
INTERLEAVED	0.107	0.299
SEPARATE	0.088	0.287

When expanding our delay beyond a fixed delay of 1, our model performance depreciates (see ??). This is biologically intuitive, since as the delay increases the amount of intermediary distractive stimulus increases as well, which should limit the model's capacity to retain information from the sample image. Our interleaved model falls to a loss oscillating around 0.693, indicating guessing at random, with delays ≥ 5 .

When trying to abstract to experiment 3, all of our models struggled. After three epochs, all models oscillated between 0.693 ± 0.2 , indicating that the models were, broadly speaking, randomly guessing. To ameliorate this, various batch sizes (2, 4, 8, 16) and learning rates (10^{-3} , 2.5×10^{-3} , 10^{-2} , 2.5×10^{-2} , 10^{-1}) were tested, but all models returned to guessing at random regardless of hyperparameter tuning.

For our simple hybrid, the model unfortunately struggled with every task and generally returned to a loss of 0.693, which indicated that the model was guessing at random.

5. Feature Learning

To perform feature learning, we look at the results of Grad-CAM for our models, a technique for "visually explaining" our CNN decisions (see A for more). To ensure that our model decision-making is sound, we explore the results of

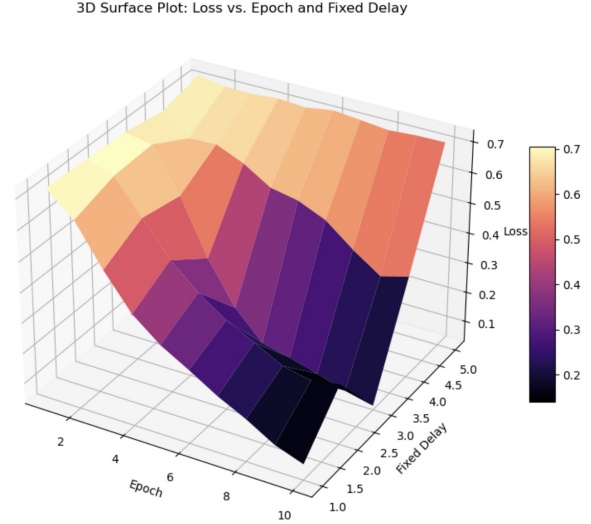


Figure 2. Meshgrid of validation losses in interleaved model for task 1, experiment 1, with differing fixed delays, after training for 10 epochs.

our interleaved and separate hybrids after training on task 2 experiment 1, with one delay frame. So that we can interpret our results, we use the PASCAL VOC 2007 dataset, which has pixel-level bounding box annotations for the object(s) contained in the image (see A for more).

After training our models to the fifth epoch, our validation loss for each model was 0.0131 and 0.0078. We calculate intersection over union (IoU) between our bounding box and our Grad-CAM results when running trials of task 2 experiment 1, which gives a mean IoU on 0.33 and 0.39 for our models. While positive, these are beneath the 0.5 threshold for IoU commonly used to indicate a positive detection. Further work is merited to identify whether this nontrivial association is due to distinctive colors in objects of our images, or more sophisticated features.

Table 3. Mean IoU of models' Grad-CAM results.

MODEL	IoU
INTERLEAVED	0.33
SEPARATE	0.39

6. Conclusion

We exhibit the working memory capacity in CNN-recurrent connectivity hybrids by proposing three hybridized models and evaluating them on a suite of DMTS tasks. We additionally explore their learned features through Grad-CAM.

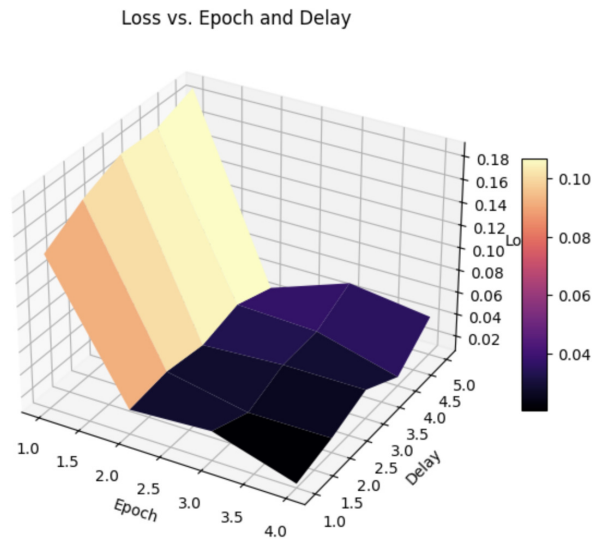


Figure 3. Meshgrid of validation losses in separate model for task 1, experiment 1, with differing fixed delays, after training for 4 epochs. Unfortunately, due to computational constraints, I was unable at what delay the model falls to guessing at random, if ever - intuitively, given the ConvLSTM layer in the separate hybrid architecture, it is possible that the model can guess above chance even with extremely long delays.

Our experiments reveal that our models exhibit better performance on distractor images than empty frames as delays, deteriorating performance as delay rises, and incapacity to scale to variable delay. The behavior of our models largely betrays what is anticipated by common models of working memory found in psychology. Future work can involve exploring architectural or more sophisticated fine-tuning mechanisms for managing variable delay tasks, and more in-depth analysis on the learned features of these models and how they are leveraged, including exploration of the representation space. Additionally, inference about biological plausibility would be greatly substantiated by an isomorphic test in animals, so that the degree of behavioral similarity can be quantified.

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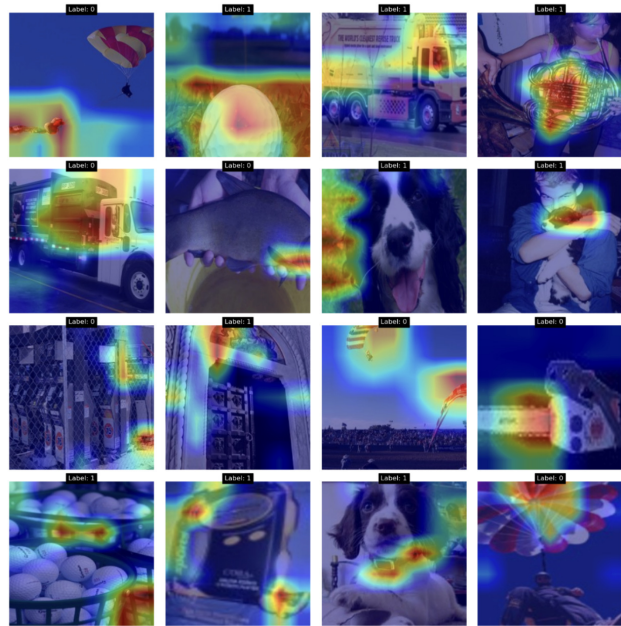


Figure 4. Example visualizations of Grad-CAM results on trials of our full hybrid in task 2 experiment 1, with one delay frame.

doi: 10.1146/annurev-psych-120710-100422. Epub 2011 Sep 27.

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A. Datasets and code

For our task experiments, we ultimately use the Imagenette dataset due to computational cost of larger datasets. Imagenette is a subset of ImageNet consisting of ten common "sysnets", or conceptual classes of images from the ImageNet dataset (tench, English springer, cassette player, chain saw, church, French horn, garbage truck, gas pump, golf ball, parachute). The dataset has a 70/30 training-validation split, which we inherit for our experiments (ima, 2019). To capture the features of the ImageNet dataset, we integrate the ResNet-50 CNN architecture, a standard CNN pretrained on the ImageNet that was introduced in (He et al., 2016).

For exploring learned representations, we use the PASCAL Visual Object Classes (VOC) 2007 dataset, which has twenty conceptual classes (vehicles, household, animals, and other: aeroplane, bicycle, boat, bus, car, motorbike, train, bottle, chair, dining table, potted plant, sofa, TV/monitor, bird, cat, cow, dog, horse, sheep, and person). Each image has pixel-level bounding box annotations. The dataset has 1,464 training images and 1,449 validation images, which we inherit for our experiments (pas, 2007). We additionally use the TensorFlow `transforms.Normalize()` function to bring our images to the same color distribution of ImageNet, in hopes that the learned features of ResNet-50 on ImageNet remain informative (for the whole ImageNet dataset, the mean pixel colors are [0.485, 0.456, 0.406] and standard deviation is [0.229, 0.224, 0.225]). We also utilize Gradient-weighted Class Activation Mapping (Grad-CAM), which was introduced in (Selvaraju et al., 2016). Grad-CAM "uses the gradients of any target concept, flowing into the final convolutional layer to produce a coarse localization map highlighting important regions in the image for predicting the concept" - in practice we use this to explore which regions of images are informative to our models' label. In actually encoding Grad-CAM, we implement the code base from (Gil, 2025). I additionally explored activation maximization, and though this was not fruitful felt I should credit the code base here.

Experiments have cumulatively taken around 200 hours to run.

Our code is available [here](#). Our experiments were run with one node and eight cores with the Harvard Open OnDemand platform.

B. Models and tasks

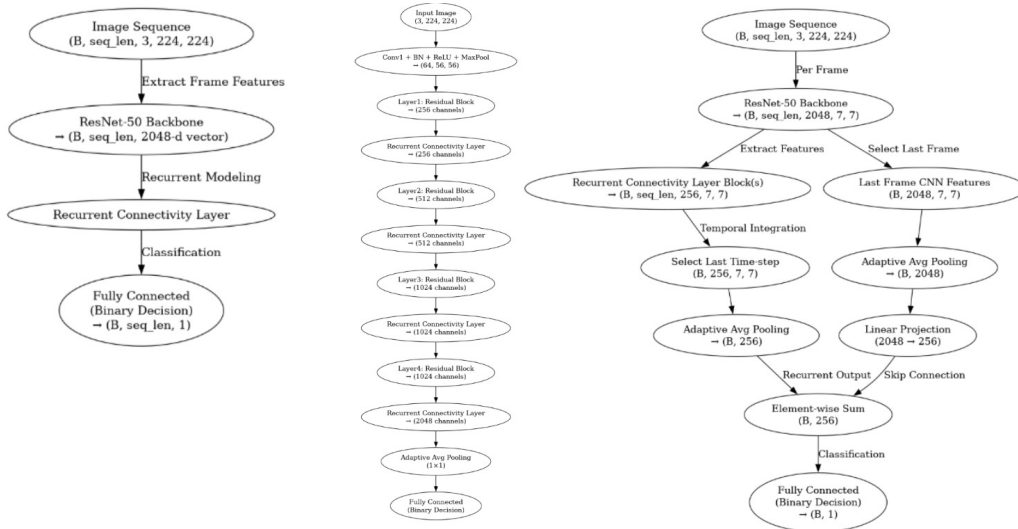


Figure 5. Visualization of our simple hybrid (left), our interleaved hybrid (middle), and our separate hybrid (right)

ResNet-50, introduced in (He et al., 2016), has (as the name suggests) 50 convolutional layers. The architecture has an initial convolution layer and final fully-connected layer. Conventionally, the remaining 48 layers are grouped into triplets

to constitute 16 residual blocks. Of these blocks, they are typically subdivided into four layer stages, with three, four, six, and three residual blocks. The number of input channels of each stage rise from 64, to 128, to 256, to 512. I use these conventions when proposing my models. More information can be found [here](#).

The ConvLSTM layer, introduced in (Shi et al., 2015), provides a simple mechanism for high-accuracy nowcasting in spatiotemporal sequence forecasting problems, and allows for storing visual features of images through an arbitrarily-long sequence of delay frames.

Our tasks were at default trained with a batch size of 32 for our simple and separate hybrids; for the interleaved hybrid, memory constraints limited the batch size to 8. Our learning rate was initialized at 10^{-3} .

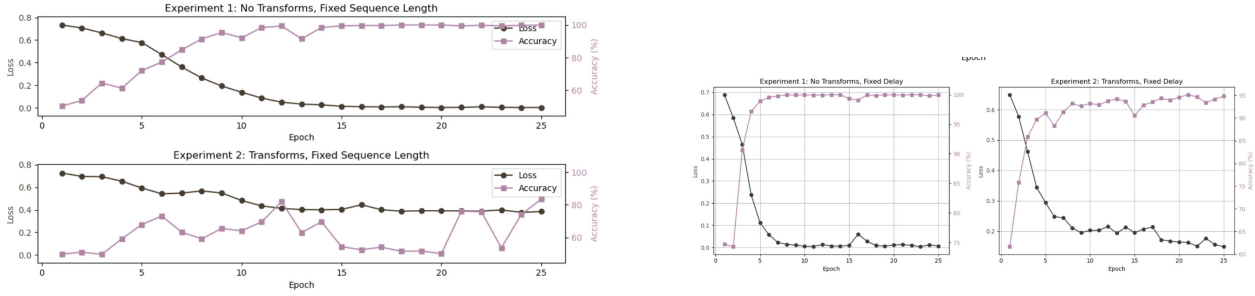


Figure 6. Our interleaved hybrid’s losses and accuracies in task 1, experiments 1 and 2 (left) and task 2, experiments 1 and 2 (right)

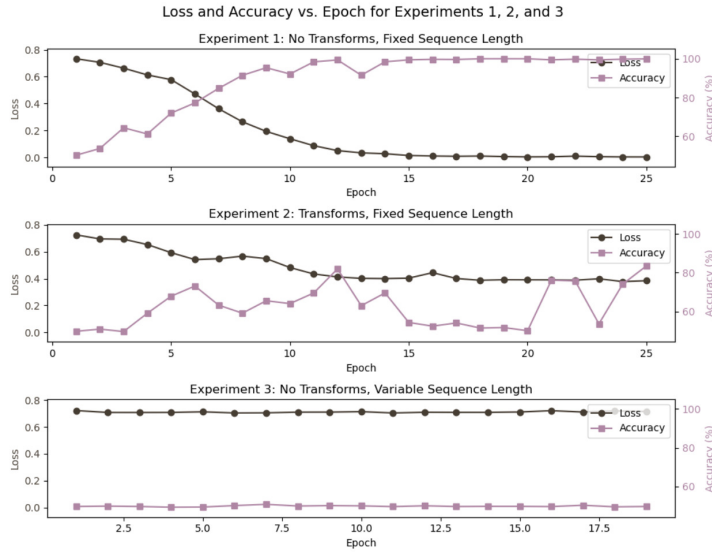


Figure 7. Our simple hybrid’s losses and accuracies in task 1, experiments 1 through 3

C. Acknowledgement

I want to thank the NEURO 140/240 team for providing me with direction and assistance throughout this project (this is the final report for this class, after all!). I am particularly thankful to Professor Kreiman, as well as to teaching fellows Xiaolong Luo and Purab Seth for meeting with me during office hours to help guide my project and navigate challenges. This project has taken me from misunderstanding machine learning fundamentals to an integrated outlook on machine learning and cognitive science and a new interest in researching what it means for humans and models to think.

The simple and interleaved hybrids, as well as tasks 1 and 2, were ideated in the NEURO 140/240 prospectus document for a project on working memory. This report is written using the template for ICLR reports, found [here](#).